

Fruit and Vegetable Intake, Physical Activity, and Mortality in Older Community-Dwelling Women



OBJECTIVES To examine the relationship between fruit and vegetable intake, physical activity, and all-cause mortality in older women. **DESIGN** Six Cox proportional hazards models examined independent and additive relationships between physical activity, carotenoids, and all-cause mortality. Additional models tested whether physical activity and carotenoids were conjointly related to mortality. Models were adjusted for age, education, and race and ethnicity. **SETTING** Baltimore, Maryland. **PARTICIPANTS** Seven hundred thirteen women aged 70 to 79 participating in the Women's Health and Aging Studies. **MEASUREMENTS** Total serum carotenoids, a marker of fruit and vegetable intake, and physical activity were measured at baseline. Physical activity was measured according to kilocalorie expenditure. **RESULTS** During 5 years of follow-up, 82 (11.5%) participants died. Measured continuously, physical activity improved survival (HR = 0.52, 95% CI = 0.41–0.66, $P < .001$). The most active women were more likely to survive than the least physically active women (HR = 0.28, 95% CI = 0.13–0.59, $P < .001$). Continuous measures of carotenoids improved survival (HR = 0.67, 95% CI = 0.51–0.89, $P = .01$). Women in the highest tertile of total carotenoids were more likely to survive those in the lowest (HR = 0.50, 95% CI = 0.27–0.91, $P = .03$). When examined in the same model, continuous measures of physical activity (HR = 0.54, 95% CI = 0.42–0.68, $P < .001$) and carotenoids (HR = 0.76, 95% CI = 0.59–0.98, $P = .04$) predicted survival during follow-up. **CONCLUSION** The combination of low total serum carotenoids and low physical activity, both modifiable risk factors, strongly predicted earlier mortality. These findings provide preliminary support that higher fruit and vegetable intake and exercise improve survival.